

Saswata Jana

Research Scholar (Prime Minister's Research Fellow) · Department of Mathematics
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RESEARCH INTERESTS

Distributed algorithms for mobile robots and agents (mutual visibility, gathering, partitioning); graph algorithms and combinatorial optimization; approximation algorithms; pursuit–evasion and graph protection games; vehicle and drone routing and scheduling.

EDUCATION

Ph.D. in Mathematics — Indian Institute of Technology Guwahati 2021 – Present

Advisor: Prof. Partha Sarathi Mandal.

Thesis Title: Miles to Walk Before Delivery: An Approximation (Thesis submitted on 22.07.26).

M.Sc. in Mathematics — Indian Institute of Technology Patna 2019 – 2021

CPI: 9.67 / 10. Institute Silver Medal and Aryabhata Gold Medal.

B.Sc. in Mathematics — Asutosh College, University of Calcutta 2016 – 2019

First Class with a percentage 67%.

PUBLICATIONS

Refereed Journal Articles

- [J1] Subhajit Pramanick, **Saswata Jana**, Partha Sarathi Mandal. *Fault-tolerant mutual visibility without any axis agreement in presence of mobility failure*. Theoretical Computer Science (Elsevier), Vol. 1024, February 2025.
- [J2] Subhajit Pramanick, **Saswata Jana**, Adri Bhattacharya, Partha Sarathi Mandal. *Mutual visibility of luminous robots despite angular inaccuracy*. Theoretical Computer Science (Elsevier), Vol. 1011, October 2024.
- [J3] **Saswata Jana**, Partha Sarathi Mandal. *Approximation algorithms for drone delivery scheduling with a fixed number of drones*. Theoretical Computer Science (Elsevier), Vol. 991, April 2024.

Refereed Conference Proceedings

- [C1] Subhajit Pramanick, **Saswata Jana**, Partha Sarathi Mandal. *Asynchronous fault-tolerant mutual visibility*. Proc. 33rd Int. Colloquium on Structural Information and Communication Complexity (SIROCCO 2026), LNCS - 16488.
- [C2] Prajyot Pyati, Navjot Kaur, **Saswata Jana**, Adri Bhattacharya, Partha Sarathi Mandal. *Separation of unconscious robots with obstructed visibility*. Proc. 22nd Int. Conference on Distributed Computing and Intelligent Technology (ICDCIT 2026), LNCS - 16420.
- [C3] **Saswata Jana**, Subhajit Pramanick, Adri Bhattacharya, Partha Sarathi Mandal. *Time-optimal asynchronous minimal vertex covering by myopic robots on graphs*. Proc. 21st Int. Symp. on Algorithmics of Wireless Networks (ALGOSENSORS 2025), LNCS -16078.
- [C4] **Saswata Jana**, Giuseppe F. Italiano, Partha Sarathi Mandal. *Graph traversal via connected mobile agents*. Proc. 21st Int. Symp. on Algorithmics of Wireless Networks (ALGOSENSORS 2025), LNCS-16078.
- [C5] **Saswata Jana**, Giuseppe F. Italiano, Manas Jyoti Kashyop, Athanasios L. Konstantinidis, Evangelos Kosinas, Partha Sarathi Mandal. *Online drone scheduling for last-mile delivery*. Proc. 31st Int. Colloquium on Structural Information and Communication Complexity (SIROCCO 2024), LNCS -14662.
- [C6] Subhajit Pramanick, **Saswata Jana**, Adri Bhattacharya, Partha Sarathi Mandal. *Distributed uniform partitioning of a region using opaque async luminous mobile robots*. Proc. 25th Int. Conf. on Distributed Computing and Networking (ICDCN 2024), ACM.
- [C7] Subhajit Pramanick, **Saswata Jana**, Adri Bhattacharya, Partha Sarathi Mandal. *Mutual Visibility with ASYNC Luminous Robots Having Inaccurate Movements*. Proc. 19th Int. Symp. on Algorithmics of Wireless Networks (ALGOSENSORS 2023), LNCS -14061.

- [C8] **Saswata Jana**, Partha Sarathi Mandal. *Approximation algorithms for drone delivery scheduling problem*. Proc. of 11th International Conference on NETworked sYStems (NETYS 2023), LNCS-14067.
- [C9] **Saswata Jana**, Partha Sarathi Mandal. *Approximation algorithms for drone delivery packing problem*. Proc. of 24th Int. Conference on Distributed Computing and Networking (ICDCN 2023), ACM.

Under Review / Submitted

- [S1] **Saswata Jana**, Partha Sarathi Mandal. *Approximating the energy-constrained drone delivery packing problem for last-mile logistics*. Submitted to Discrete Mathematics.
- [S2] Subhajit Pramanick, **Saswata Jana**, Gokarna Sharma, Partha Sarathi Mandal. *Asynchronous gathering of opaque robots with mobility faults*. Submitted to 28th Int. Symp. on Stabilization, Safety, and Security of Distributed Systems (SSS 2026).
- [S3] Prajyot Pyati, Navjot Kaur, **Saswata Jana**, Adri Bhattacharya, Partha Sarathi Mandal. *Separation of unconscious robots with obstructed visibility*. Submitted to Theoretical Computer Science (Elsevier).
- [S4] **Saswata Jana**, Giuseppe F. Italiano, Manas Jyoti Kashyop, Athanasios L. Konstantinidis, Evangelos Kosinas, Partha Sarathi Mandal. *Online drone scheduling for last-mile delivery*. Submitted to Theoretical Computer Science (Elsevier).

PROJECT EXPERIENCE

Master's Project — Vizing's Conjecture and VC Dimension of Graphs July 2020 – May 2021
Guide: Dr. Subhabrata Paul, IIT Patna.

AWARDS AND FELLOWSHIPS

Prime Minister's Research Fellowship (PMRF), Government of India July 2022 – July 2026
Aryabhata Gold Medal, IIT Patna 2021
Institute Silver Medal, IIT Patna 2021
INSPIRE Scholarship, DST, Government of India 2016 – 2021
SVMCM Scholarship, Government of West Bengal 2014 – 2016
Qualified GATE and IIT JAM 2019, 2021

TEACHING EXPERIENCE

Teaching Assistant, IIT Guwahati

- Computer Programming (MA511), M.Sc., Jul–Nov 2022
- Data Structures Laboratory (in C) (MA619), Ph.D., Jan–May 2023
- Data Structure Laboratory (in C/ C++) MA252), (MA512), M.Sc., Jan–May 2024 and 2025
- Multivariate Calculus (MA101), Btech, Jul–Dec 2025
- Data Structures Laboratory (in C+) (MA252), Btech, Jul–Dec 2025

Instructor, North Gauhati College

- Mathematical Modeling, Jan–Apr 2023 and 2024
- Numerical Methods, Jul–Sep 2023
- Number Theory, Oct–Dec 2023, 2024 and 2026
- Automata Theory, Jan–Apr 2025

TECHNICAL SKILLS AND LANGUAGES

Programming: C, C++, Python, LaTeX.
Languages: Bengali (native), English, Hindi.